

EE2080: Random Processes

Programming Assignment Instructions

Overview

- **Questions:** 3 (all mandatory)
- **Total Marks:** 30
- **Duration:** 1.5 hours
- **Environment:** Local Python installation, coding live in an IDE
- **Recording:** Entire screen must be recorded with system time visible at all times

Setup

1. Use a computer with a local Python installation.
2. Open your preferred IDE
3. Prepare a folder named: **PA_<RollNo>_<Name>** for all source files.
4. Launch a full-screen recorder to capture the entire screen.
5. Ensure the system time/clock is visible on-screen at all times.
6. Close all non-essential applications and turn off notifications.

Start of Exam

1. Start screen recording **before** you begin coding.
2. Show the empty working folder on screen, then begin coding.

During the Exam

1. Code live in your IDE using the local Python environment.
2. Do not switch to other applications (browsers, chat apps, file explorers beyond the assignment folder, etc.).
3. **No AI tools are allowed.** This includes ChatGPT, Copilot, Gemini, etc.
4. No external resources (web searches, forums, or pre-written code).
5. Keep the system time visible throughout the session.
6. Save files frequently using filenames: **<RollNo>_Q1.py, <RollNo>_Q2.py, <RollNo>_Q3.py.**
7. **Write clean, readable code** with comments where necessary.
8. **If using terminal commands,** ensure they are visible in the recording.
9. Debug independently without external help.

End of Exam

1. Stop coding exactly at 1.5 hours.
2. Run each solution file and show outputs on the recording.
3. Show folder contents and file timestamps.
4. Stop the recording only after demonstrations are completed.
5. Submit the files in the assignment on Google Classroom as instructed previously.

EE2080: Random Processes

Programming Assignment - Set:1

23 August, 2025

1 Moving Average Process (5 Marks)

Consider a random process $\{X_n : n \in \mathbb{N}\}$ where $X_n \sim \mathcal{N}(0, 1)$, i.i.d.

Define the moving average process

$$Y_n = \frac{1}{8} \sum_{i=0}^7 X_{n-i}, \quad n \in \{11, 12, \dots, 20\}.$$

Simulate $M = 10^3$ realizations and:

- Estimate and plot the mean process $\mu_Y(n)$, for $n \in \{11, 12, \dots, 20\}$.
- Estimate the covariance $C_Y(\Delta) = \frac{1}{M} \sum_{i=1}^M y_{n_1}^i y_{n_2}^i$, using the realizations, for all $\Delta = n_2 - n_1$ in valid range.

exph.

2 Gambler's Ruin (10 Marks)

A gambler starts with wealth k dollars and has a target $n > k$. At each step t he chooses a stake b_t from $\{2, 3\}$, subject to affordability ($b_t \leq$ his current wealth), according to a given betting policy. If he bets \$2, he wins with probability p_1 and if he bets \$3, he wins with probability p_2 . For a given (k, n, p_1, p_2) -tuple, consider the following betting strategies:

- Always-1: Bet 2\$ always
- Threshold: Bet \$3 if wealth $W_t \geq T$, else bet \$2. Set the threshold T at 10\$ as default.

Tasks:

- Plot two sample paths for each betting policy
- Histogram of stopping time per policy
- Threshold sweep: Sweep $T \in \{2, 4, 6, \dots, 20\}$ and plot the success probability vs T and the average stopping time vs T .
- For a given (k, n, p_1, p_2) -tuple and the default value of T , output the estimated success probabilities and average stopping-times of both the betting policies and declare the better policy w.r.t success probability.

Default value of $(k, n, p_1, p_2) = (10, 30, 0.3, 0.7)$

3 Hitting Times and Mean Recurrence Times (15 marks)

Consider a time homogeneous discrete-time Markov chain with state space

$$S = \{1, 2, 3, 4, 5, 6\}$$

and the transition probability matrix

$$P = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0.2 & 0 & 0.8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 0.4 & 0.6 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 0.3 & 0.7 & 0 \end{bmatrix}$$

1. **Hitting time simulation.** Let $T_{i \rightarrow A}$ denote the first hitting time of a set A starting from state i . Here, take $A = \{6\}$ and write code to simulate the chain starting at state 2 to estimate

$$h_{2,6} \triangleq \mathbb{E}[T_{2 \rightarrow \{6\}}].$$

Run at least 10^5 independent sample paths and report the empirical mean.

2. **Empirical mean recurrence time.** Run a single long sample path, (a.k.a trajectory) (e.g., 10^6 steps) of the chain and estimate the average return time to state 5:

$$\hat{m}_2 = \frac{1}{K} \sum_{k=1}^K (\tau_k - \tau_{k-1}),$$

where $0 = \tau_0 < \tau_1 < \dots < \tau_K$ are successive visit times to state 5.

3. **Stationary distribution and mean recurrence times.** Compute the stationary distribution π by solving

$$\pi P = \pi, \quad \sum_{i \in S} \pi_i = 1, \quad \pi_i \geq 0.$$

$$\begin{aligned} \pi(P - I) &= 0 \\ (P^T - I) \pi^T &= 0 \\ \left[\begin{array}{c} P^T - I \\ \sum \pi_i \end{array} \right] \left[\begin{array}{c} P^T - I \\ 1 \end{array} \right] \pi^T &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}$$